19 f and g are functions such that

$$f(x) = \frac{12}{\sqrt{x}} \quad \text{and} \quad g(x) = 3(2x + 1)$$

(a) Find g(5)

$g(5) = 3(2 \times 5 + 1) = 33.$

1 mark
33

(1)

(b) Find gf(9)

$f(9) = 12/\sqrt{9} = 12/3 = 4.$

$gf(9) = g(4) = 3(2 \times 4 + 1) = 27.$

2 marks total
f(9) = 4 then g(4)
27

(2)

(c) Find $g^{-1}(6)$

$g(x) = 6$ means $3(2x + 1) = 6.$

$2x + 1 = 2$, so $x = 1/2.$

2 marks total
Solve g(x)=6
x = 1/2

(2)

(Total for Question 19 is 5 marks)

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